EE5320: Analog Integrated Circuit Design
Assignments

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January-May 2026

Assignment 1

Use the square law model with $\lambda = 0\,(g_{ds} = 0)$ for MOS transistors unless otherwise mentioned.

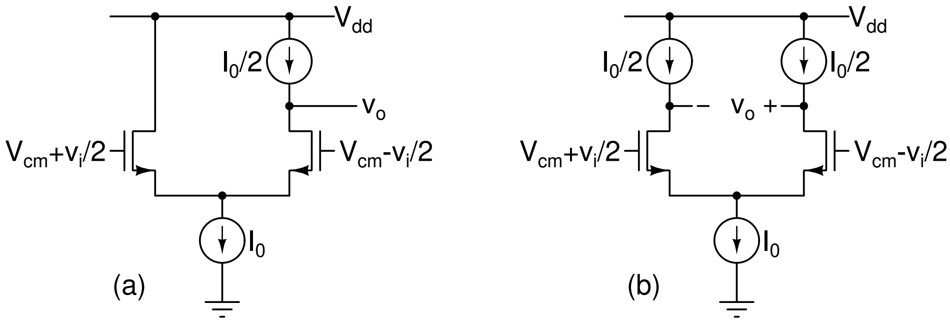


Figure 1.1: Problem 1.1.

1.1. Determine v_o/v_i in Fig. 1.1(a) and (b). Model the MOS transistors using their g_m and g_{ds}. Explain the results very briefly.

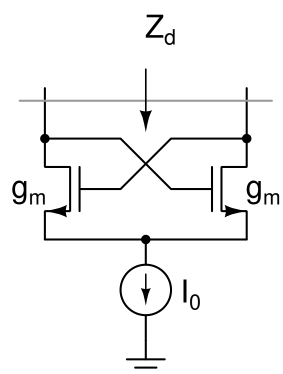


Figure 1.2: Problem 1.2.

1.2. Determine Z_d in the circuit in Fig. 1.2.

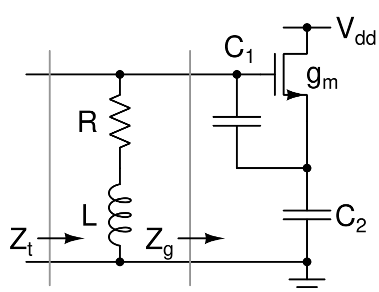


Figure 1.3: Problem 1.3.

1.3. Determine $Z_g(j\omega)$ in the circuit in Fig. 1.3. Determine ω_o, L and R such that $Z_t(j\omega_o) = \infty$.

1.4. Determine $Z_g(j\omega)$ in the circuit in Fig. 1.4. Determine L_g such that $Z_i(j\omega_o)$ has zero imaginary part. What is $Z_i(j\omega_o)$ in that case?

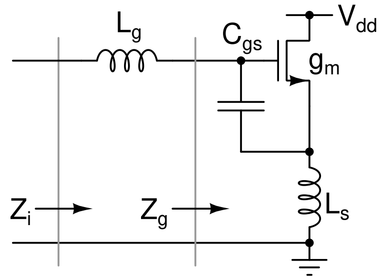


Figure 1.4: Problem 1.4.

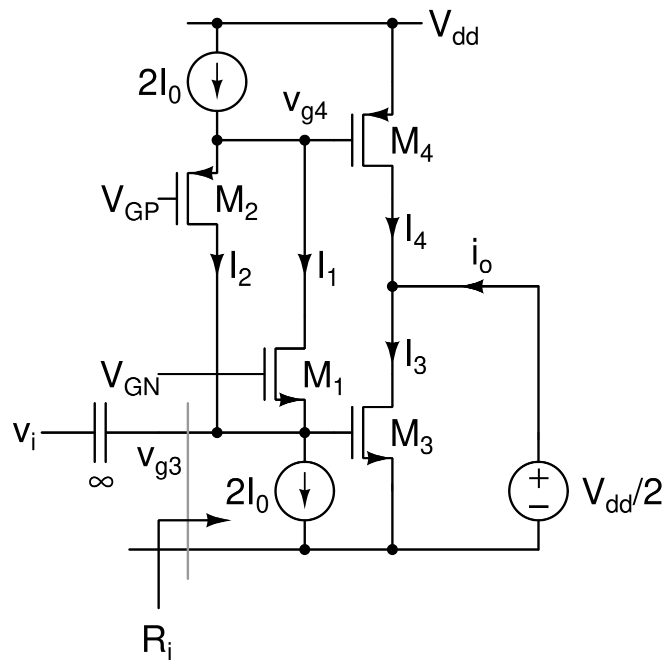


Figure 1.5: Problem 1.5.

1.5. Aspect ratios of $M_{1,2}$ are W/L and those of $M_{3,4}$ are nW/L. Quiescent drain currents of $M_{1,2}$ are I_0 and those of $M_{3,4}$ are nI_0. All transistors are in the saturation region. Determine V_{GP} and V_{GN}. Determine the small-signal input resistance R_i. Determine the small-signal voltages v_{g3} and v_{g4} when a small signal v_i is applied as shown. Determine i_o/v_i. The answers for small-signal quantities should be in terms of I_0 and transistor dimensions.

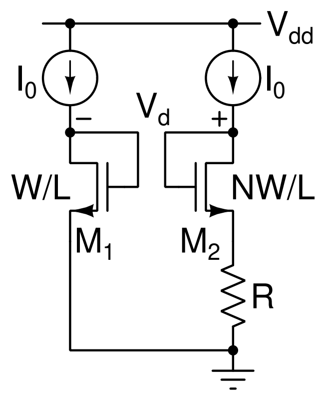


Figure 1.6: Problem 1.6.

1.6. Use the square law model for the MOS transistors. Determine V_d as a function of I_0. What is I_0 (non-zero solution) when $V_d = 0$? What is the g_m of M_1 in this condition? Sketch V_d versus I_0. Axes and key values must be clearly marked.

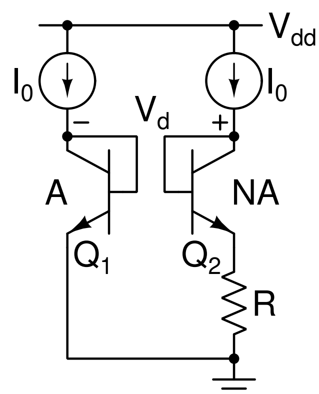


Figure 1.7: Problem 1.7.

1.7. Use the exponential relationship for the BJTs. $I_c = AJ_s \exp(V_{BE}/V_t)$ where J_s is the reverse saturation current density and A is the area. Determine V_d as a function of I_0. What is I_0 (non-zero solution) when $V_d = 0$? What is the g_m of Q_1 in this condition? Sketch V_d versus I_0. Axes and key values must be clearly marked.

1.8. Model the MOS transistors in Fig. 1.8 with their g_m, C_{gs}, and C_{db}.

(a) Determine the transfer function V_o/V_s and its -3 dB bandwidth (rad/s) in Fig. 1.8(a). Repeat the same for Fig. 1.8(d-f) with $R_G = 0$.

(b) Fig. 1.8(b, c) show L added to Fig. 1.8(a). Determine the transfer function V_o/V_s.

(c) Determine L for the highest bandwidth without peaking (see below). Determine the new -3 dB bandwidth as a multiple of $1/RC_L$.

(d) Repeat the same for Fig. 1.8(d-f) with $R_G \neq 0$. Determine R_G for the highest bandwidth without peaking (see below).

(e) In Fig. 1.8(f), Determine the new -3 dB bandwidth (numerically) for two cases: $C_o = 0.5C, C_i = 0.5C$ and $C_o = 0.75C, C_i = 0.25C$ where C_o is the capacitance at the output node and C_i is the capacitance at the other end of R_g.

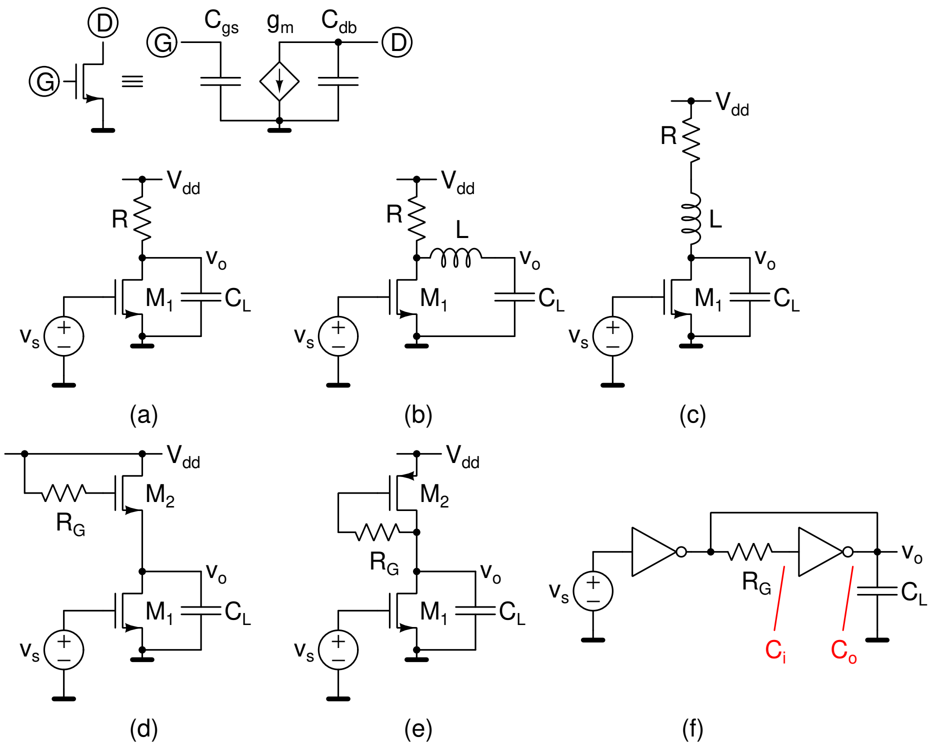


Figure 1.8: Problem 1.8.

Consider the N^{th} order low-pass transfer function ($M < N$)

$$H(s) = A_o \frac{1 + b_1 s + b_2 s^2 + \ldots + b_M s^M}{1 + a_1 s + a_2 s^2 + \ldots + a_N s^N}$$

The maximally flat condition for the above is given by equating the coefficients in the numerator and denominator of normalized $|H(j\omega)|^2$. i.e., make $d_k = c_k$, $1 \le k \le M$ and $d_k = 0$, $M + 1 \le k \le N - 1$.

$$|H(j\omega)|^2 = A_o^2 \frac{1 + c_1\omega^2 + c_2\omega^4 + \ldots + c_M\omega^{2M}}{1 + d_1\omega^2 + d_2\omega^4 + \ldots + d_N\omega^{2N}}$$

Assignment 2

Use the square law model with $\lambda = 0\,(g_{ds} = 0)$ for MOS transistors unless otherwise mentioned.

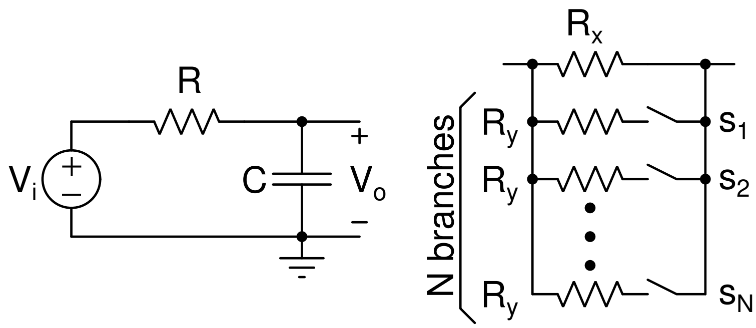


Figure 2.1: Problem 2.1.

2.1. The resistors and capacitors in a CMOS process vary by ±25% and ±15% respectively across process corners. The RC filter in Fig. 2.1 is designed for a nominal bandwidth of 10 MHz.

(a) Determine the maximum and minimum bandwidths across process corners.

(b) The resistor is realized as a combination of a fixed resistor R_x and an array of N equal resistors R_y that can be switched in parallel. If the bandwidth has to be kept within 2.5% of the nominal bandwidth across all corners, determine nominal values of R_x, R_y, and N assuming a 10 pF capacitor (nominal value).

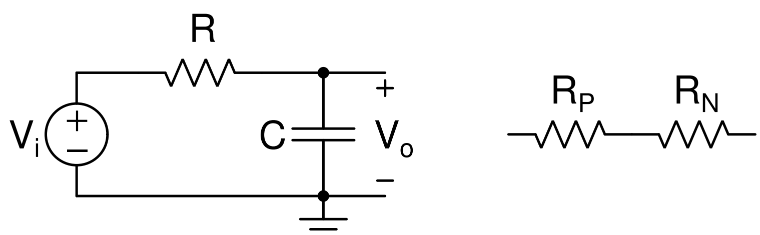


Figure 2.2: Problem 2.2.

2.2. Fig. 2.2 shows a 10 MHz bandwidth RC filter with a 10 pF capacitor. The resistor R is realized as a series combination of R_P and R_N which have a positive and negative temperature coefficients respectively in an attempt to have the bandwidth invariant with temperature. The sheet resistances (at room temperature) and temperature coefficients are as follows: R_P: 100 Ω/sq., +0.001/°C; R_N: 150 Ω/sq., −0.002/°C

The temperature dependence is modeled as $R(T) = R(T_0)(1 + \alpha(T - T_0))$ where T_0 is the room temperature and α is the temperature coefficient. Determine the following such that the bandwidth is independent of temperature:

(a) Room temperature resistance values $R_P(T_0)$ and $R_N(T_0)$ and their Aspect ratios L_P/W_P and L_N/W_N.

(b) Maximum and minimum bandwidths in the −40 to 100°C range if only the positive temperature coefficient resistor (with the correct value at 27°C) is used.

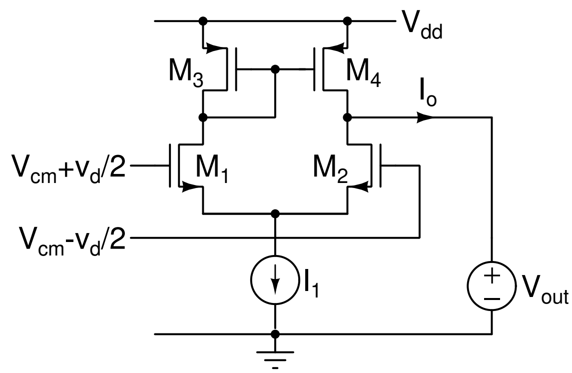


Figure 2.3: Problem 2.3.

2.3. Fig. 2.3 shows a differential pair. All transistors are in the saturation region. Neglect λ. Determine the following:

(a) Output current I_o with $v_d = 0$ due to mismatch. The answer must be in terms of individual mismatch values ΔV_{T1-4}, $(\Delta\beta/\beta)_{1-4}$, and transistor g_ms.

(b) Standard deviation σ_{Io} of the output mismatch current in terms of A_{VTN}, $A_{\beta N}$, A_{VTP}, $A_{\beta P}$, transistor dimensions W_{1-4}, L_{1-4}, and transistor g_ms.

(c) Input voltage v_d that results in $I_o = 0$. The answer must be in terms of individual mismatch values ΔV_{T1-4}, $(\Delta\beta/\beta)_{1-4}$, and transistor small signal parameters.

(d) Standard deviation of the above v_d in terms of A_{VTN}, $A_{\beta N}$, A_{VTP}, $A_{\beta P}$, transistor dimensions W_{1-4}, L_{1-4}, and transistor g_ms.

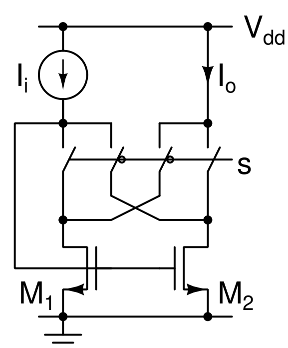


Figure 2.4: Problem 2.4.

2.4. In Fig. 2.4, the switches are controlled by the signal s. The switches without the bubble are on and off for $s = 1$ and $s = 0$ respectively. The opposite is true for the switches with the bubble. The two transistors have the same W, L. They are in the saturation region. Neglect λ. Sketch the circuit for $s = 1$ and $s = 0$. Determine the following:

(a) Standard deviation of the current gain I_o/I_i when s is fixed at 1.

(b) Determine W and L if $I_o = 500\mu A$, $V_{GS} - V_T = 200\,mV$, and the standard deviation of the current gain is 1%. $A_{VT} = 5\,mV\,\mu m$, $A_\beta = 1\,\%\,\mu m$. Nominal values: $\mu_n C_{ox} = 400\,\mu A/V^2$, $V_{TN} = 0.5\,V$.

(c) s switches between 0 and 1 with a 50% duty cycle. Sketch $I_o(t)$ for one cycle of s. Determine the time average of $I_o(t)$ over one period and its standard deviation.

The answers must be in terms of A_{VTN}, $A_{\beta N}$, transistor dimensions W, L, and transistor g_ms.

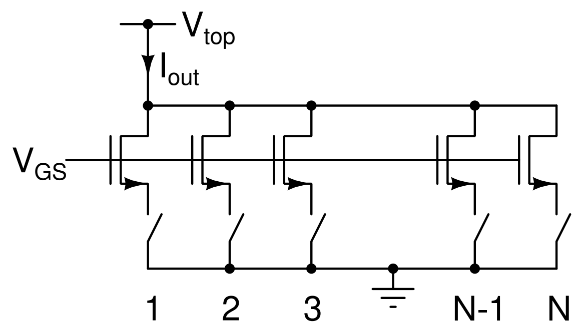


Figure 2.5: Problem 2.5.

2.5. Fig. 2.5 shows N identical MOS transistors controlled by switches. Each transistor carries a nominal current I_o and has an overdrive voltage $V_{GS} - V_T$. The switches are successively turned on. The output current is denoted by $I_{out}[n]$ when the switches 1 to n are turned on. $0 \le n \le N$. Determine the following:

(a) Mean and standard deviation of $I_{out}[n]/I_{avg}$ where $I_{avg} = I_{out}[N]/N$. The answer should be in terms of A_{VT}, A_β, N, transistor dimensions W, L and transistor g_m. For what n does it reach the maximum? What is the maximum value?

(b) Mean and standard deviation of $(I_{out}[n+1] - I_{out}[n])/I_{avg}$.

(c) Mean and standard deviation of $(I_{out}[2n+1] - 2I_{out}[n])/I_{avg}$. Briefly comment on this and the previous result.

Hint: To keep the expressions compact, derive the expressions in terms of the total current error in the unit source $\Delta I_o/I_o$ or its standard deviation. You can separately give the expression for $\sigma(\Delta I_o/I_o)$ in terms of the transistor's mismatch parameters.

Mismatch is relatively small. Use approximations such as $(1+x)(1+y) \approx 1 + x + y$ where applicable.

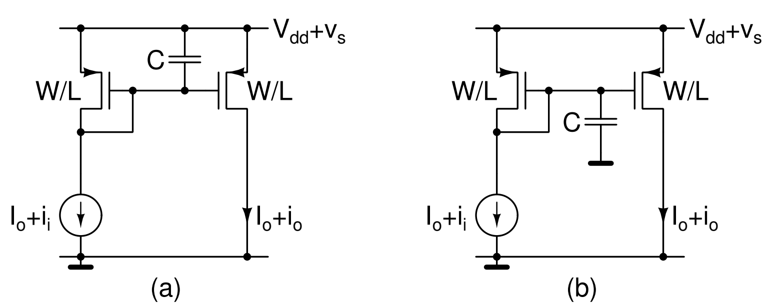


Figure 2.6: Problem 2.6.

2.6. Determine the small signal transfer functions $I_o(s)/I_i(s)$ and $I_o(s)/V_s(s)$ in Fig. 2.6(a) and Fig. 2.6(b). What do you infer from this? Model the transistors using their g_m.

Assignment 3

Use the square law model with $\lambda = 0\,(g_{ds} = 0)$ for MOS transistors unless otherwise mentioned.

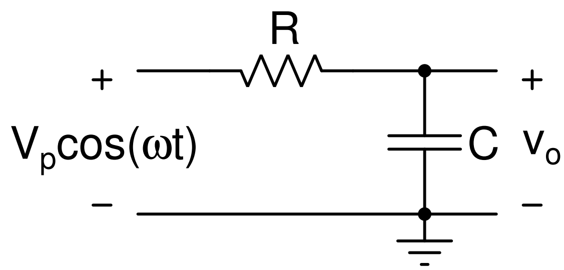


Figure 3.1: Problem 3.1.

3.1. The filter in Fig. 3.1 is driven by a sinusoid at $\omega = 1/RC$. Determine the following:

(a) Output signal to noise ratio *SNR* (ratio of mean squared signal to mean squared noise voltages), and the power dissipated in the circuit. .

(b) Expression for power dissipated in the circuit in terms of the *SNR* above and the bandwidth $f_B = 1/2\pi RC$. What tradeoffs does this relationship represent?

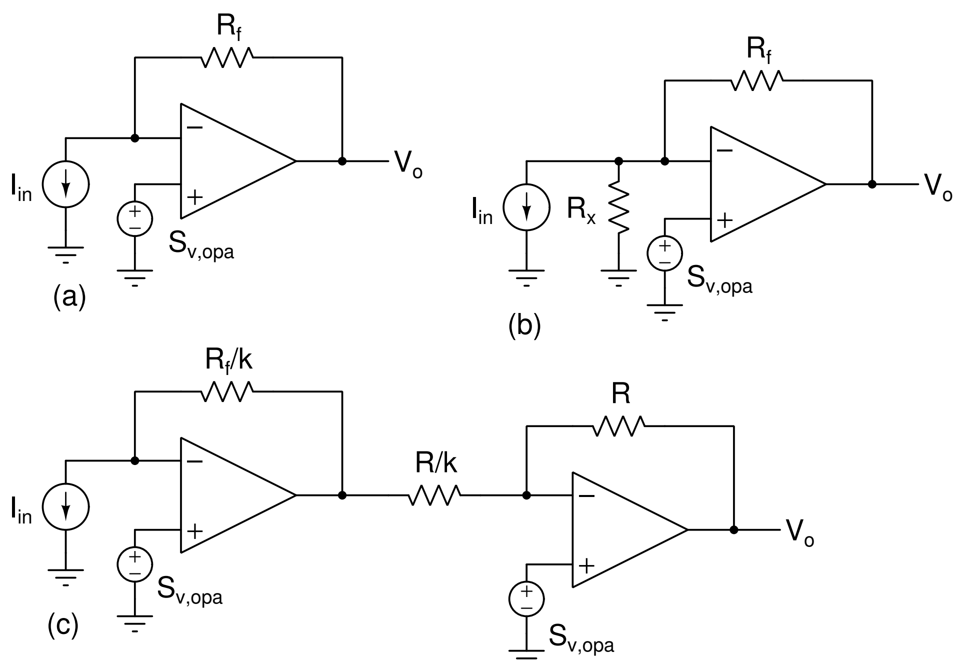


Figure 3.2: Problem 3.2: (a) Single-stage, (b) Single-stage with additional conductance between the virtual ground and ground, (c) Two-stage.

3.2. Fig. 3.2 shows three different transmpendance amplifier configurations. Determine the gain V_o/I_{in} and the input referred noise (referred to I_{in}) in these circuits. Briefly comment on the results. $S_{v,opa}$ is the input referred voltage noise spectral density of the opamp.

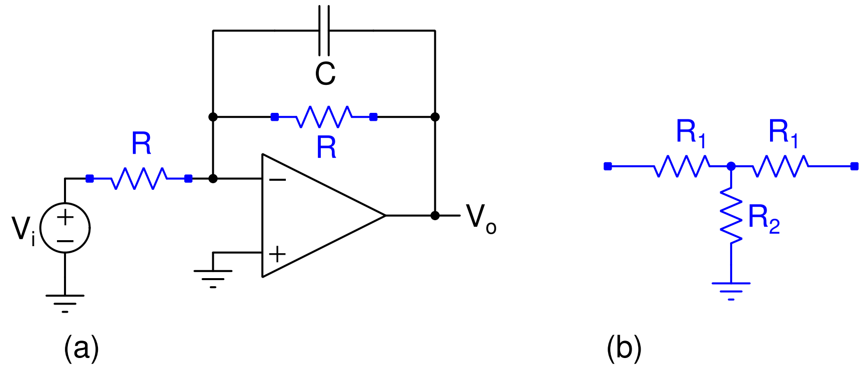


Figure 3.3: Problem 3.3.

3.3. Fig. 3.3(a) shows an RC low-pass filter with a bandwidth of 10^4 rad/s. $C = 1$ pF.

(a) Determine R, the input referred noise spectral density (in V^2/Hz), and the rms output noise (in µV). What is the total resistance required?

(b) Because of area constraints, the total resistance that can be used on the chip is restricted to 2 MΩ. The structure in Fig. 3.3 is used in place of the two resistors. Determine R_1 and R_2 so that the transfer function remains the same while using the maximum allowed total resistance(Hint: Determine y_{12} of the network in Fig. 3.3 and equate it to that of the single resistor.).

(c) Determine the input referred noise spectral density (in V^2/Hz), and the rms output noise (in µV). Comment on the results.

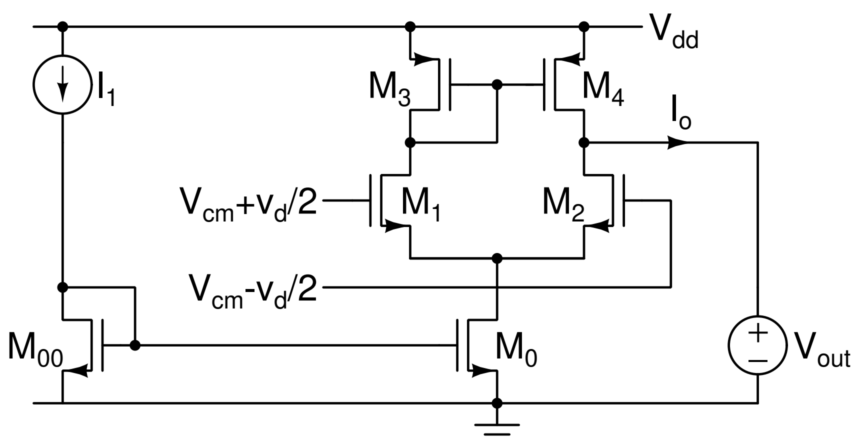


Figure 3.4: Problem 3.4.

3.4. Fig. 3.4 shows a differential pair. All transistors are in the saturation region. $M_{00,0}$, $M_{1,2}$, and $M_{3,4}$ are matched pairs. Determine the following:

(a) Spectral density of the output current noise I_o. The answer must be in terms of transistor g_ms. What are the contributions of M_{00} and M_0?

(b) Spectral density of the input referred noise voltage (referred to v_d). The answer must be in terms of transistor g_ms.

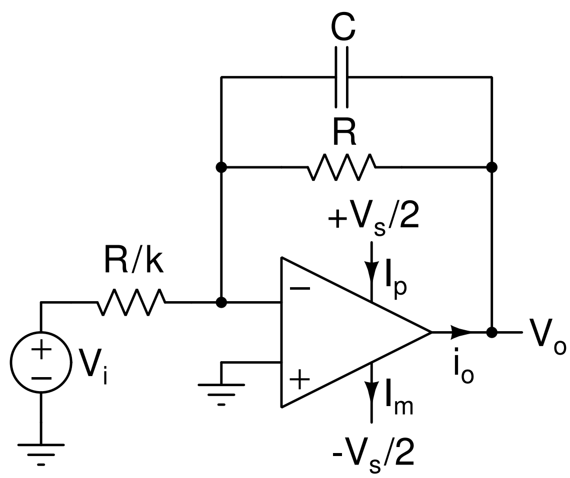


Figure 3.5: Problem 3.5.

(Not for submission) Fig. 3.5 shows an RC low-pass filter with a dc gain k. It is driven by an input $V_i = (V_{pp}/k)\cos(t/RC)$. The opamp can be either Class-A or Class-B type. The supply currents in either case are as follows.

3.5.
- Class-A: $i_p = I_{bias}$, $i_p - i_m = i_o$
- Class-B: If $i_o > 0$, $i_p = i_o$ and $i_m = 0$. If $i_o < 0$, $i_p = 0$ and $i_m = -i_o$.

Determine the following (for both opamps above). Compare the results to those of Problem 3.1.

(a) Output signal to noise ratio *SNR* (ratio of mean squared signal to mean squared noise voltages), and the average power drawn from the supplies (given by $(V_s/2)\left(\overline{i_p} + \overline{i_n}\right)$). The overbar denotes the average over one period.

(b) Expression for power dissipated in the opamp in terms of the *SNR* above and the bandwidth $f_B = 1/2\pi RC$. Arrange the relationship to show the gain k and the ratio V_{pp}/V_s. What tradeoffs does this relationship represent?

3.6. (a) Determine the g_m of M_1 in Fig. 3.6(a) so that the input source is matched, i.e., $R_{in} = R_s$. Determine the gain v_o/v_s.

(b) Determine the input-referred thermal noise PSD due to M_1 and R_s. The answer should be in terms of R_s.

(c) Determine the g_m of M_1 in Fig. 3.6(b) so that the input source is matched, i.e., $R_{in} = R_s$. Determine the gain v_o/v_s. V_{G0} is adjusted such that M_1 and M_2 have the same V_{GS}.

(d) Determine the input-referred thermal noise PSD due to M_1, M_2 and R_s. The answer should be in terms of R_s.

(e) Determine the g_m of M_1 (M_1 and M_2 are identical) in Fig. 3.6(c) so that the input source is matched, i.e., $R_{in} = R_s$. R_{bias} is very large. Determine the gain $v_o/2v_s$. $v_o = v_{op} - v_{om}$.

(f) Determine the input-referred (referred to v_s) thermal noise PSD due to M_1, M_2, and R_s. The answer should be in terms of R_s.

(g) Not for submission. Repeat the above for Fig. 3.6(d, e).

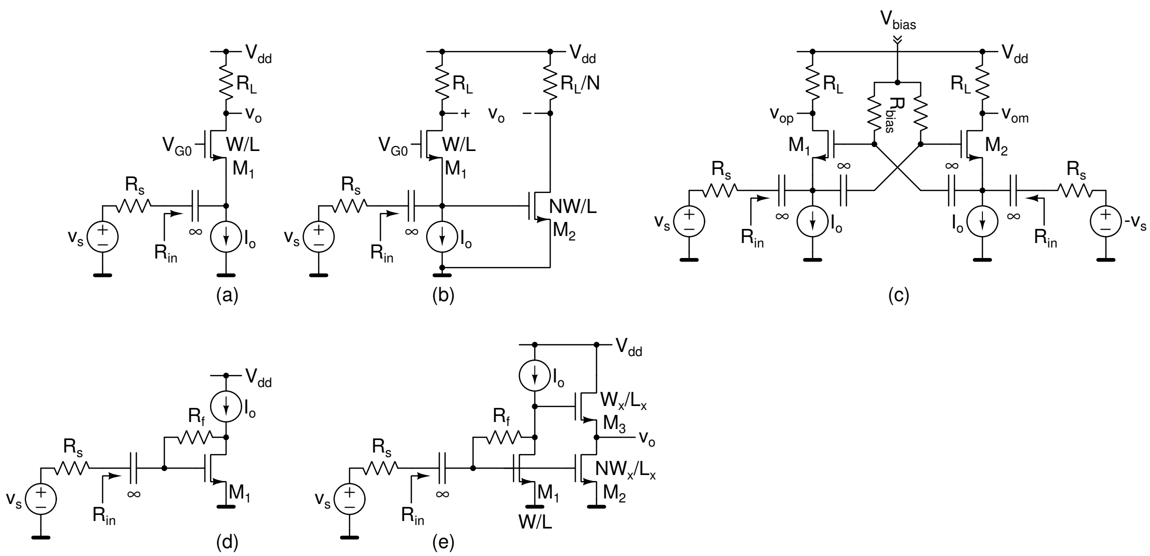


Figure 3.6: Problem 3.6.

Assignment 4

Use the square law model with $\lambda = 0\,(g_{ds} = 0)$ for MOS transistors unless otherwise mentioned.

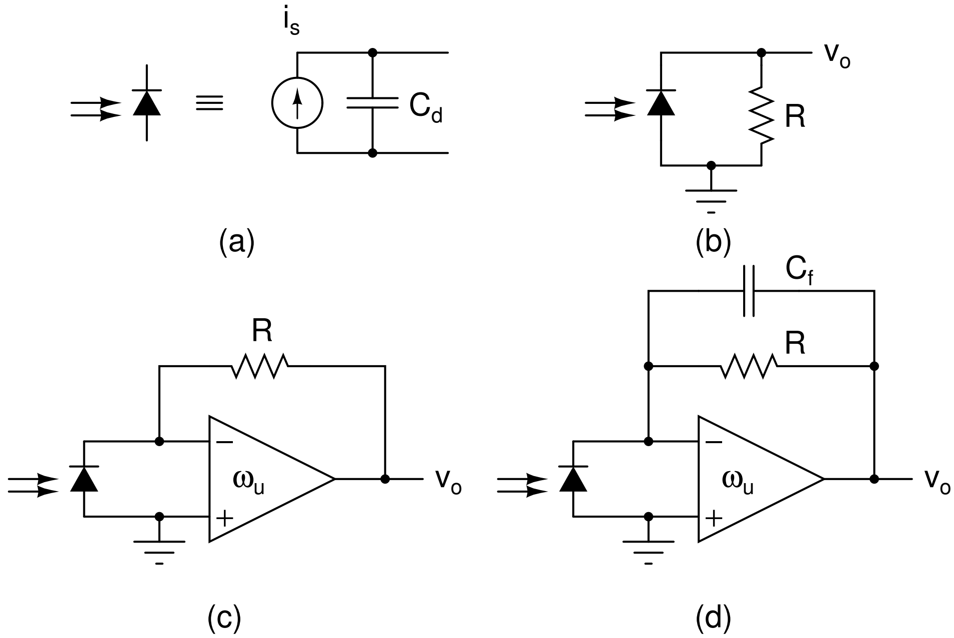


Figure 4.1: Problem 4.1.

4.1. Fig. 4.1 shows the model for a photodiode. The diode is setup in reverse bias (biasing arrangement not shown) and when light shines on it, a current is produced. The current can be converted to voltage using the schemes in Fig. 4.1(b) to (d). This circuit is used in optical communication systems. The goal is to maximize the gain as well as the bandwidth of conversion from i_s to v_o. Do the following for a given R (dc gain V_o/I_s).

   (a) What is the bandwidth in Fig. 4.1(b)? What is the gain bandwidth product?

   (b) What is the bandwidth in Fig. 4.1(c)? Choose the opamp's ω_u such that the quality factor of the denominator polynomial is $1/\sqrt{2}$ (Same as damping factor $\zeta = 1/\sqrt{2}$). With this quality factor, you get a maximally flat magnitude response (maximum possible bandwidth without peaking; derivatives $d^n/d\omega^n|H(j\omega)|^2$ are zero to the maximum degree n that is possible.). What is the gain bandwidth product? What are the advantages and disaadvantages of Fig. 4.1(c) when compared to Fig. 4.1(b)? (One sentence each)

   (c) What is the bandwidth in Fig. 4.1(c)? Assume an ω_u for the opamp and then choose C_f such that you get a maximally flat magnitude response. What is the gain bandwidth product? What are the advantages and disaadvantages of Fig. 4.1(d) when compared to Fig. 4.1(b, c)? (One sentence each)

4.2. Fig. 4.2 shows a unity gain amplifier with a capacitive load. The transconductor G_m has input and output capacitors $C_i = C_o = G_m/\omega_T$ where ω_T is a technology parameter. Assume that realizing G_m requires a bias current $I_b = 0.5(V_{GS} - V_T) \cdot G_m$. Treat $V_{GS} - V_T$ as a constant in this problem (it depends on the internal details of the transconductor). Determine the transfer function of the amplifier. Determine the bias current required to have the transfer function pole at a certain frequency ω_B. Plot I_b versus ω_B. What is your inference from this plot? If the transconductor's noise can be represented by an output noise current with a PSD $4kTG_m$, determine the integrated output noise as a function of ω_B.

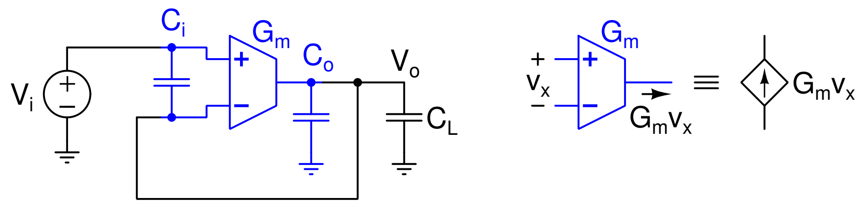


Figure 4.2: Problem 4.2.

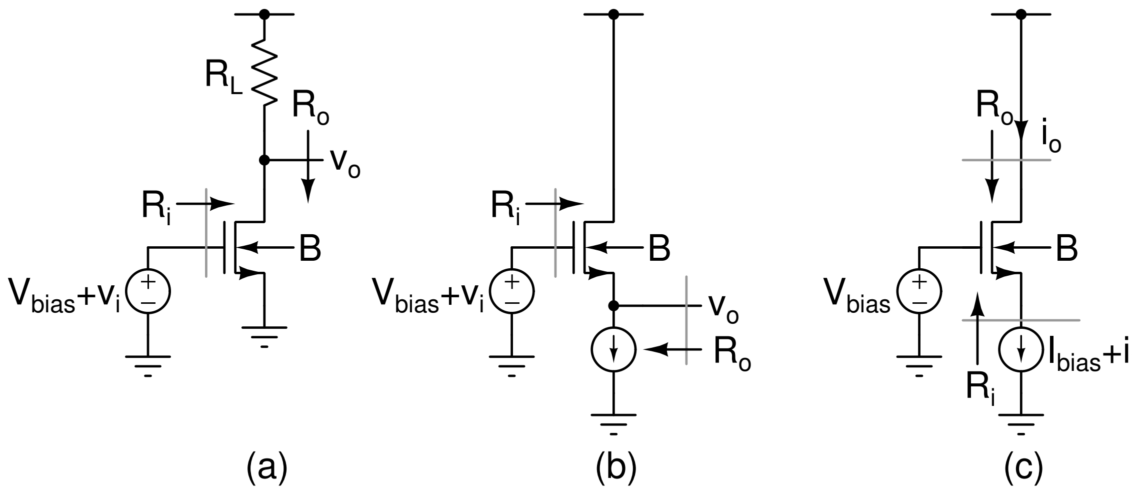


Figure 4.3: Problem 4.3. (a) Common-source amplifier, (b) Common-drain amplifier, (c) Common-gate amplifier. Bulk terminal "B" is connected to either the transistor's source or ground.

4.3. Fig. 4.3 shows the basic single-transistor amplifier stages. Determine the gain (v_o/v_i or i_o/i_i as applicable), input resistance, and the output resistance when (a) the bulk terminal "B" is connected to the source, and (b) when it is connected to ground. Show the small signal equivalent circuits and very brief calculations. Present the final results in the form of a table with these quantities for the two cases. Very briefly comment on whether body effect makes them "better" or "worse". Model the transistors using g_m, g_{ds}, and g_{mb}.

4.4. Fig. 4.4 shows the basic cascode amplifier and a gain-boosted cascode amplifier. Give the expressions for the output impedance Z_o in these cases. In case of the gain-boosted opamp, show the results with (a) $A(s) = \omega_u/s$ and (b) $A(s) = A_o/(1 + s/p_1)$, $p_1 = \omega_u/A_o$. Sketch the Bode magnitude and phase plots of Z_o, on the same set of axes for all 3 cases. What is the utility of the gain-boosted cascode?

4.5. The loop gain $L(s)$ of a system with N extra poles and $M < N$ extra zeros is given by

$$L(s) = \frac{\omega_{u,loop}}{s} \frac{\sum_{m=0}^{M} b_m s^m}{\sum_{n=0}^{N} a_n s^n}$$

$b_0 = a_0 = 1$. What does the loop gain step response (inverse laplace transform of $L(s)/s$) look like after the initial transient period? Give your answer in terms of the poles of the additional factor (Hint: Split $L(s)$ into a sum of two parts, one of which is $\omega_{u,loop}/s$; This problem doesn't require a lot of algebra, but requires reasoning and basics of polynomials)

4.6. Determine the closed loop natural frequency and damping factor of an opamp in unity feedback when the opamp

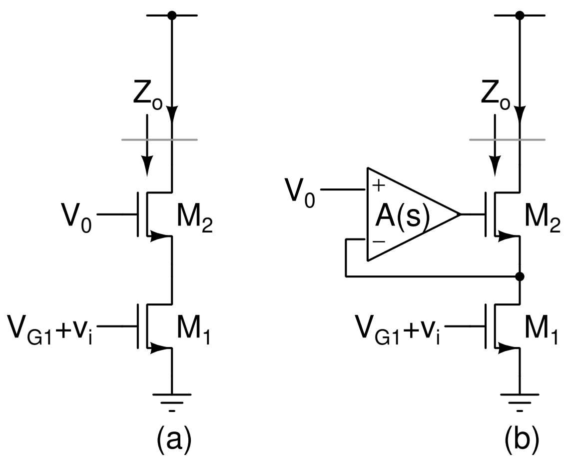


Figure 4.4: Problem 4.4. (a) Cascode, (b) Gain boosted cascode.

gain if A_1 or A_2 given below. Also determine the condition for critical damping

$$A_1(s) = \frac{\omega_u}{s}\frac{1}{1+s/p_2}$$

$$A_2(s) = \frac{\omega_u z_1}{s^2}\left(1+\frac{s}{z_1}\right)$$

Sketch the loop gain in the two cases (for critically damped closed loop) on the same axes. Mark key points clearly.

Assignment 5

Use the square law model with $\lambda = 0\,(g_{ds} = 0)$ for MOS transistors unless otherwise mentioned.

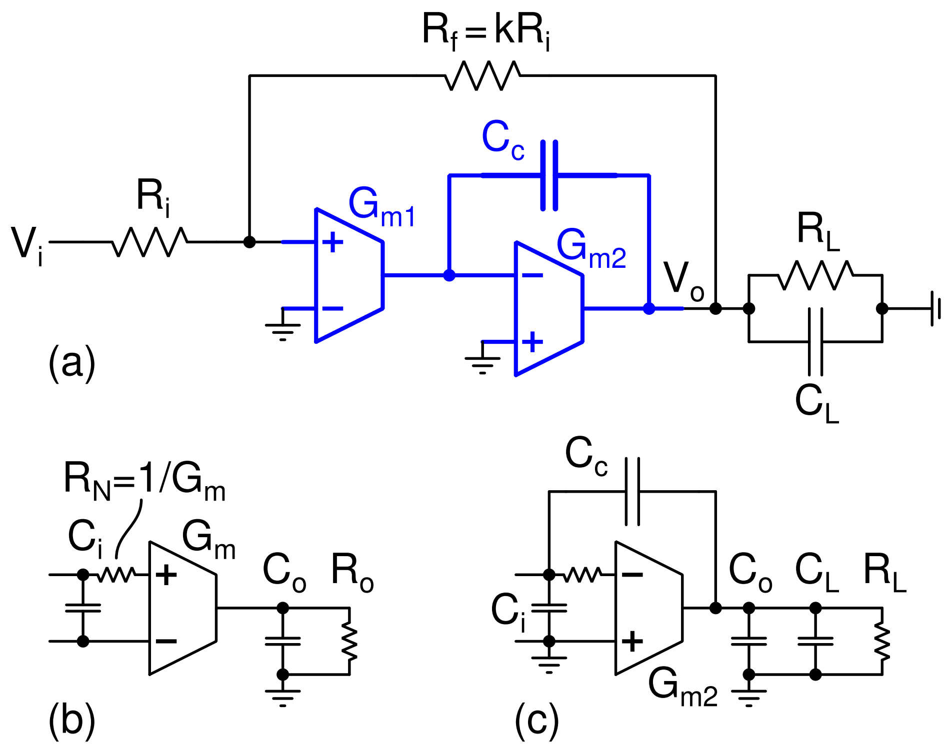


Figure 5.1: Problem 5.1.

5.1. Fig. 5.1(a) shows an inverting amplifier of gain $-k$ using a two-stage Miller-compensated opamp (shown in blue). The specifications are given below.

Specification	Value	Unit
Closed loop dc gain k	8 + last digit of your roll number	—
Closed loop bandwidth f_B	20 − last digit of your roll number	MHz
Load capacitor C_L	4 + mod (last two digits of your roll number, 25)	pF
Load resistor R_L	$1/(\pi f_B C_L)$	Ω
Input resistance R_i	10 + last digit of your roll number	kΩ

You should design this amplifier using controlled sources for $G_{m1,2}$. Use $C_c = 1$ pF (limited by area). The non-dominant pole p_2 must be at four times the unity loop gain frequency $\omega_{u,loop}$.

Model each transconductor as shown in Fig. 5.1(b). $C_i = C_o = G_m/\omega_T$, $\omega_T = 2 \times 10^{10}$ rad/s, $G_o = G_m/100$. $R_N = 1/G_m$ to model the thermal noise.

Go about the amplifier design step-by-step as suggested below.

(a) Use the network shown in Fig. 5.1(c) to determine G_{m2} from the constraint on the non-dominant pole. This network consists only of G_m and its associated parasitics, G_L, C_L, and C_c. You will get a quadratic equation for G_{m2}. You can neglect the output conductance of G_{m2} at this step.

(b) Determine G_{m1}.

(c) Add all the components to realize the closed loop amplifier. Each transconductor muse be modeled as shown in Fig. 5.1(b).

Simulate the amplifier with the above determined values and present the following results:

(a) Table showing the specification table with your specific values and all the component values.

(b) Loop gain magnitude (dB) and phase (degrees). The unity loop gain frequency and phase margin must be marked. Break the loop at the input of G_{m1} to simulate the loop gain. The frequency range should be from $\sim 0.1\times$ dominant pole to where the loop gain is ~ -20 dB.

(c) Closed loop transfer function magnitude on a log-y scale showing the dc gain and the 3-dB bandwidth.

(d) Unit step response.

(e) Output noise PSD and the input referred noise PSD.

(f) Table showing dc gain, 3-dB bandwidth, unity loop gain frequency, phase margin, input noise PSD at low frequencies, fraction (in percent) of noise contributions to low frequency output noise PSD from R_i, R_f, G_{m1}, G_{m2}, and R_L.

(Not for submission): Connect a resistor $R_c = 1/G_{m2}$ in series with C_c. Simulate the loop gain, closed loop transfer function, and step response. Compare with the earlier design.

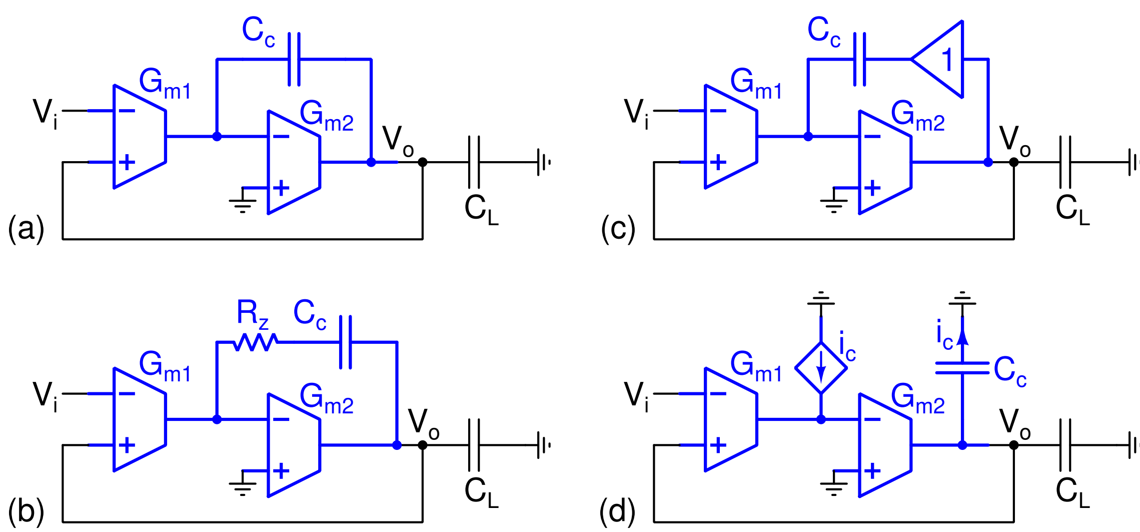


Figure 5.2: Problem 5.2: (a) Conventional Miller opamp, (b) With R_c in series with C_c, (c) With a buffer to drive C_c, (d) With a CCCS to complete the feedback around the second stage.

5.2. Use the previously designed opamp (with the same C_L) to design the voltage follower in Fig. 5.2(a). Simulate the phase margin. (To simulate the phase margin, you can break the loop at the input of G_{m1} and add the input capacitance of G_{m1} to C_L.)

Connect R_c in series with C_c as shown in Fig. 5.2(b). Sweep the value of R_c from zero to $10/G_{m2}$ and find the R_c for optimum phase margin. How does this compare to the phase margin in Fig. 5.2(a)?

Use an ideal unity gain buffer to drive C_c as shown in Fig. 5.2(c). Determine the phase margin. How does this compare to the phase margin in Fig. 5.2(a)?

Use a CCCS to drive the current into the input of the first stage as shown in Fig. 5.2(d). Determine the phase margin. How does this compare to the phase margin in Fig. 5.2(a)?

(You should show a table with phase margin values in each of the cases. You should also show the transfer function of the opamps—denote the total capacitance at the outputs of G_{m1} and G_{m2} as C_1 and C_2 respectively.)

Assignment 6

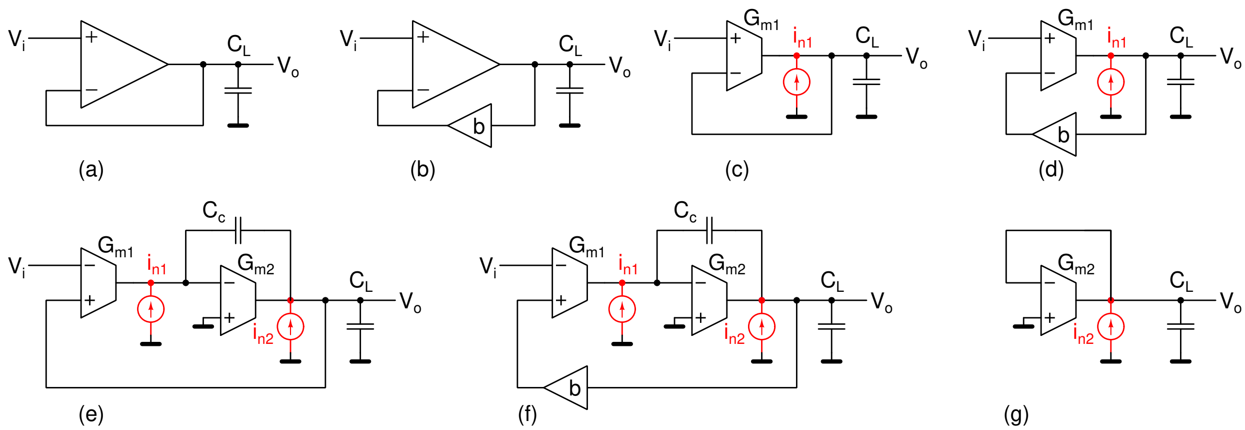


Figure 6.1: Problem 6.1. (a) Opamp in unity feedback with load C_L, (b) Opamp with a feedback fraction $b = 1/A_o$ and a load C_L, (c, d) Single-stage opamp realization of (a, b), (e, f) Two-stage Miller opamp realization of (a, b).

6.1. Fig. 6.1(a) shows a unity gain opamp-based buffer with a load C_L. Fig. 6.1(b) shows an opamp with a feedback fraction b to realize a dc gain $A_o = 1/b$. Fig. 6.1(c-f) shows realizations of Fig. 6.1(a,b) using single-stage and two-stage opamps. In each of them, calculate the following. The noise of a transconductance G_m can be modeled using an output noise current with a power spectral density (PSD) $4kTG_m$ or an input noise voltage with a power spectral density $4kT/G_m$. For calculating the integrated noise refer to the handout titled "Contour integration to calculate integrals."

- Input-referred noise PSD of the circuits in Fig. 6.1(c-f). Express the noise PSD as the input-referred noise PSD of G_{m1} and an additional factor that has the dc gain A_o, and capacitor or G_m ratios.
- DC gain, unity loop gain frequency $\omega_{u,loop}$ and parasitic poles and zeros, p_2 and z_1 (if applicable).
- Integrated output noise of the circuits in Fig. 6.1(c-f). Express the noise PSD as the product of kT/C_L and additional factors that has the dc gain A_o and capacitor or G_m ratios.
- Noise bandwidth of the circuits in Fig. 6.1(c-f). The noise bandwidth is the ratio of integrated output noise divided by the product of input referred noise spectral density and squared dc gain.
- Express the input referred noise PSD in terms of "specifications" unity loop gain frequency $\omega_{u,loop}$, load C_L, dc gain A_o, and "design choices" $a_c = C_c/C_L$ and $a_2 = p_2/\omega_{u,loop}$.
- What is the integrated output noise in Fig. 6.1(g)? How is it related to the contribution of G_{m2} in Fig. 6.1(e, f).

6.2. Determine the output noise current spectral density for the differential pair with a current mirror load and for the folded-cascode opamp. Assume that all transistors have the same gate overdrive. For the folded-cascode opamp, assume that the current in each of the output mirror branches is the same as the current in each of the differential pair transistors.

Assignment 7

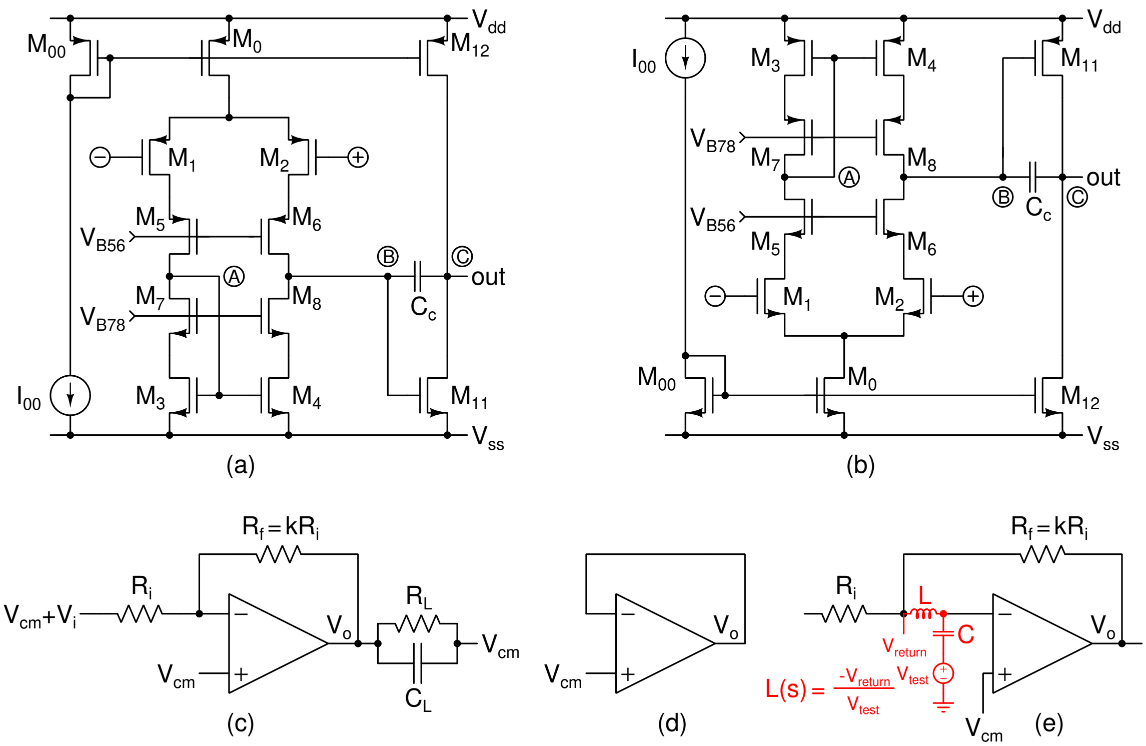


Figure 7.1: Problem 7.1.

7.1. Fig. 7.1 shows two-stage opamps. The first stage is a telescopic cascode stage. (a) uses a pMOS differential pair in the first stage and an nMOS common-source amplifier in the second stage. (b) uses an nMOS differential pair in the first stage and an pMOS common-source amplifier in the second stage.

Realize the inverting amplifier (Fig. 7.1(c)) that you designed in assignment 4 using these opamps. You will be using the same component values here. You will also need the results of MOS characterization to determine the current density.

For simulating the operating point, place the opamp in unity negative feedback as shown in Fig. 7.1(d) or simulate it in open loop with nodes A, B, and C shorted to each other and both inputs at V_{cm}.

- $V_{dd} = 1.8$ V, $V_{ss} = 0$ V, $V_{cm} = 0.9$ V.
- Simulation temperature: 100°C.
- Odd roll numbers: pMOS input pair; Even roll numbers: nMOS input pair.
- $L = 0.3\,\mu$m for all transistors.
- Use[1] $V_{DSAT} = 0.15$ V for the first stage transistors except M_0. Use $V_{DSAT} = 0.25$ V for the second stage transistors, M_0, and M_{00}.
- Choosing the V_{DSAT} value fixes the g_m and I_D per $1\,\mu\text{m}/0.3\,\mu\text{m}$ finger. Choose the multiplier m for each transistor to set its g_m or I_D as required.
- For M_{00}, use $m = 2$ and choose I_{00} (integer μA) that sets the desired bias in the first and the second stages.

[1]Choose the current density that sets the V_{DSAT} to within 10 mV of the specified values. Do not aim for microvolt precision.

- All nMOS bulk terminals should be connected to V_{ss}. All pMOS bulk terminals should be connected to V_{dd}.
- Adjust V_{B56} such that the V_{DS} of $M_{1,2}$ is 50 mV above their V_{DSAT}. Similarly, Adjust V_{B78} such that the V_{DS} of $M_{3,4}$ is 50 mV above their V_{DSAT}. i.e., M_{1-4} must be in saturation region with 50 mV margin.

Verify that the operating point (bias currents, voltages, transconductances) are correctly set up before simulating the amplifier.

Present the following results.

Tables:

(a) Specification table with your specific values and all the component values.

(b) Table showing simulation results: closed loop dc gain, closed loop 3-dB bandwidth, unity loop gain frequency, phase margin, rms output noise (integrated from 10 kHz to 100 MHz), fraction of noise variance (integrated from 10 kHz to 100 MHz) contributed by R_i, R_f, first stage, second stage, and R_L, Opamp open loop dc gain, positive and negative slew rates, positive and negative swing limits, $HD3$, supply voltage, current consumption. To find the opamp slew rates apply a large input step (from V_{cm} to $V_{cm} + V_{step}$) such that the first stage current is completely switched to one side.

To find $HD3$, apply a sinusoidal input that results in a 1 V peak-peak output at a frequency that is 1/4th the bandwidth. Report $HD3$, the ratio of the third harmonic to the fundamental (in dB).

Plots:

(a) Loop gain magnitude (dB) and phase (degrees). The unity loop gain frequency and phase margin must be marked. Break the loop as shown in Fig. 7.1(e) to simulate the loop gain. Use large L, C, e.g., $L = 10^6$ H, $C = 1$ F. The frequency range should be from $\sim 0.1\times$ dominant pole to where the loop gain is ~ -20 dB.

(b) Closed loop transfer function magnitude on a log-y scale showing the dc gain and the 3-dB bandwidth.

(c) Closed loop dc transfer curve. Vary V_i from $V_{cm} - 0.5$ V to $V_{cm} + 0.5$ V

(d) Small-signal step response: Output should step from $V_{cm} - 0.05$ V to $V_{cm} + 0.05$ V and back. Use a short rise time $\sim$ 100 ps.

(e) Large-signal step response: Output should step from $V_{cm} - 0.5$ V to $V_{cm} + 0.5$ V and back. Use a short rise time $\sim$ 100 ps. Check to see that the current is completely switched to one side in the first stage.

(f) Output noise PSD and the input referred noise PSD of the closed loop amplifier from 10 kHz to 100 MHz.

(g) Input referred noise PSD of the opamp from 10 kHz to 100 MHz.

(Not for submission):

- Remove the cascodes from the first stage, run the simulations and find out the differences
- Connect C_c to the drain of M_2 or M_4 instead of the drains of $M_{6,8}$. See if there is a difference in phase margin. You can also try connecting $C_c/2$ each to the drains of M_2 and M_4.

Assignment 8

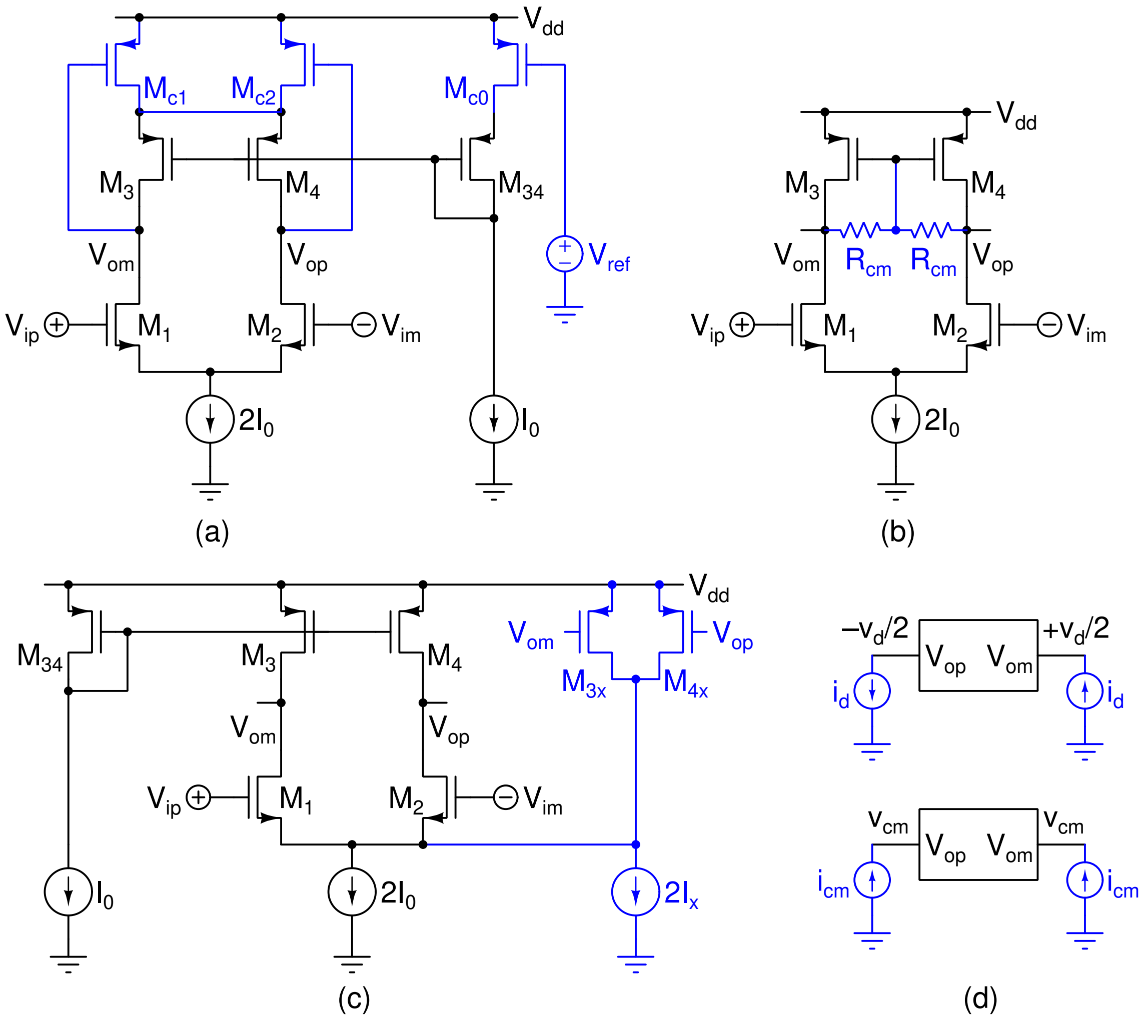


Figure 8.1: Problem 8.1. (a), (b), (c): CMFB circuits, (d) Analysing differential and common mode impedances.

8.1. Fig. 8.1(a, b, c) show three possible common-mode feedback circuits applied to a single-stage opamp. Fig. 8.1(d) shows the configuration for determining differential and common-mode resistances looking into the output terminals. Assume that all transistors are in the saturation region unless otherwise specified. Model pMOS transistors using their g_m and g_{ds}, and nMOS transistors only using g_m. Determine the following:

(a) Differential half circuit.

(b) Common-mode half circuit.

(c) Nominal output common mode voltage (operating point).

(d) CMFB loop gain (small-signal). To determine this, break the loop at the gate of a transistor.

(e) Small-signal differential output impedance $R_d = v_d/i_d$.

(f) Small-signal common-mode output resistance $R_{cm} = v_{cm}/i_{cm}$.

(g) Common-mode rejection ratio contributed by the CMFB circuit R_d/R_{cm}.

Some notes about the circuits:

(a) M_{c0} is identical to M_{c1} and M_{c2} They operate in the triode region. Use the large signal expression in the triode region and omit the V_{DS}^2 term while modeling them. V_{ref} is the common-mode reference voltage.

(b) Common-mode feedback using a resistive divider.

(c) Common-mode feedback using current injection into the tail node. M_{34} is identical to M_3 and M_4.

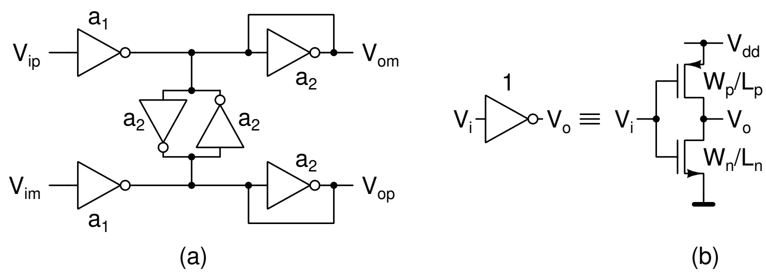


Figure 8.2: Problem 8.2. (a) Differential amplifier using a CMOS inverter, (b) Unit inverter. Numbers shown next to inverters are width-scaling factors.

8.2. Fig. 8.2(a) shows a differential amplifier using CMOS inverters. Inverters are scaled versions of the unit inverter in Fig. 8.2(b). Transistor widths are scaled by the factors shown next to the inverters. $V_{ip,im} = V_{bias} \pm v_{id}/2 + v_{icm}$. $V_{op,om} = V_{bias} \pm v_{od}/2 + v_{ocm}$. V_{bias} is the self-bias voltage of the inverter ($V_i = V_o$).

(a) Determine the small-signal differential gain v_{od}/v_{id} and common-mode gain v_{ocm}/v_{icm} of the differential amplifier. Model the inverter with its g_m and g_o. Determine the ratio a_1/a_2 to obtain a common-mode gain less than unity.

(b) Determine the current consumption of the differential amplifier as a multiple of the unit inverter's current consumption when self biased ($V_o = V_i$).

Assignment 9

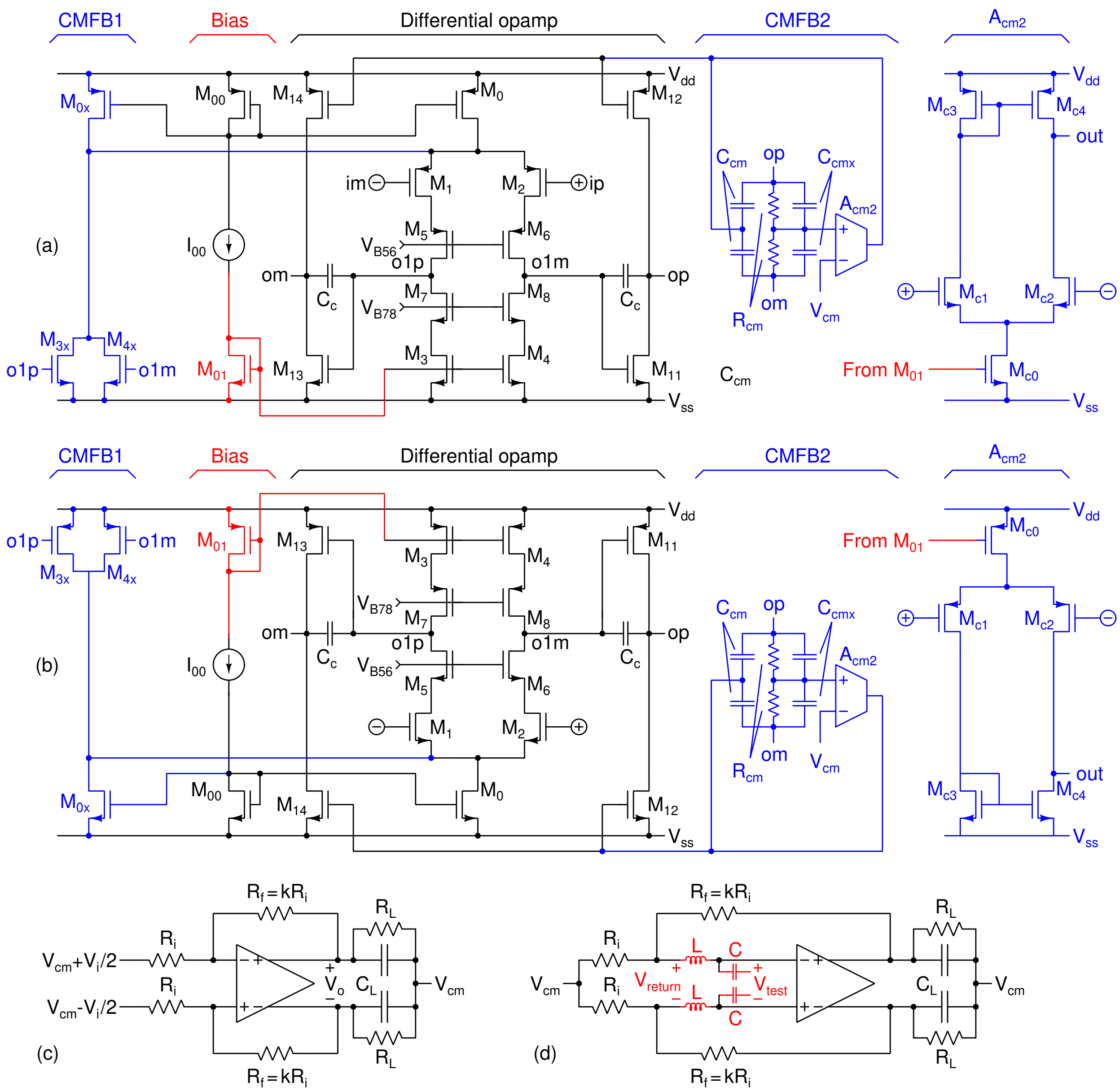


Figure 9.1: Problem 9.1. (a) Fully differential opamp with a pMOS input pair, (b) Fully differential opamp with a nMOS input pair, (c) Test circuit, (d) Loop gain test circuit.

9.1. Fig. 9.1 shows fully-differential two-stage opamps. It is a modification of the single-ended opamp from the previous assignment. Use the same bias current in the first and second stages as in the previous assignment.

- In the differential opamp, use the same transistors as in the single-ended opamp. M_{13} and M_{14} are the same as M_{11} and M_{12} respectively. V_{B56} and V_{B78} can be same as in the single-ended opamp.
- Add another transistor M_{01} to the bias branch to derive the gate bias.[2].

[2]This works fine with an ideal current source as shown here. In practice, two separate branches are required for M_{00} and M_{01}

- $M_{0x,3x,4x}$ form the common-mode feedback for the first stage. $M_{3x,4x}$ must be replicas ($\Rightarrow$ same unit transistor and current density, different number of fingers) of $M_{11,13}$. Make the current through M_{0x} the same as that through M_0. Adjust the sizes of $M_{3x,4x}$ accordingly.
- The second stage uses a linear common-mode detector using $R_{cm} = 100\,\mathrm{k\Omega}$. For A_{cm2} use the same tail current as the first stage. $V_{cm} = V_{dd}/2\,\mathrm{V}$. Initially omit C_{cm} and C_{cmx}. Step V_{cm} from $V_{dd}/2\,\mathrm{V}$ to $V_{dd}/2 + 50\,\mathrm{mV}$. The output nodes op and om should settle without ringing. If there is ringing, introduce a small C_{cm} (10 fF) and increase it until the overshoot reduces to 5%.

Realize the inverting amplifier (Fig. 9.1(c)), differential version of the opamp that you designed in assignment 4 using the opamp in Fig. 9.1(a) or (b). Use the same transistor sizes as in assignment 5 in corresponding stages. You will also need the results of MOS characterization to determine the current density.

Do the circuit design in steps. Verify proper operation (bias voltages and currents) and move to the next step.

- Construct the first stage fully differential amplifier. Bias its outputs with an ideal dc voltage source such that all transistors are in saturation. Verify the bias currents and voltages. Some residual current could be flowing through the dc voltage source, but should be small. Make sure that the tail source M_0 is in the saturation region. If it is not, use a wider (increase m) transistor to get a lower V_{DSAT}. You will also have to change M_{00} and M_{0x} appropriately.
- Construct CMFB1, connect it to the first stage outputs instead of the ideal voltage source in the above step. Verify the bias currents and voltages. The output voltage should be such that it biases the second stage at the right current.
- Add the second stage to the first. Bias the outputs op, om, with an ideal dc voltage source such that all transistors are in saturation. Verify the bias currents and voltages. Some residual current could be flowing through the dc voltage source, but should be small.
- Construct CMFB2 (without C_{cm} and C_{cmx}), connect it to the first stage outputs instead of the ideal voltage source in the above step. Verify the bias currents and voltages. The output common-mode voltage should be $V_{dd}/2$.
- Adjust C_{cm} for 5% overshoot with a common mode step as described above. Add C_{cmx} if you are unable to stabilize the loop. Use the minimum value required.
- Realize the inverting amplifier and run the required simulations.

(The transistor sizes in the first and second stages should be same as in the previous assignment. Others can be found from the current densities mentioned in the above instructions.)

- $V_{dd} = 1.8\,\mathrm{V}$, $V_{ss} = 0\,\mathrm{V}$, $V_{cm} = 0.9\,\mathrm{V}$.
- Simulation temperature: 100°C.
- Odd roll numbers: pMOS input pair; Even roll numbers: nMOS input pair.
- $L = 0.3\,\mu\mathrm{m}$ for all transistors.
- Use[3] $V_{DSAT} = 0.15\,\mathrm{V}$ for the first stage transistors except M_0. Use $V_{DSAT} = 0.25\,\mathrm{V}$ for the second stage transistors, M_0, and M_{00}.
- Choosing the V_{DSAT} value fixes the g_m and I_D per $1\,\mu\mathrm{m}/0.3\,\mu\mathrm{m}$ finger. Choose the multiplier m for each transistor to set its g_m or I_D as required.

[3] Choose the current density that sets the V_{DSAT} to within 10 mV of the specified values. Do not aim for microvolt precision.

- For M_{00}, use $m = 2$ and choose I_{00} (integer µA) that sets the desired bias in the first and the second stages.
- All nMOS bulk terminals should be connected to V_{ss}. All pMOS bulk terminals should be connected to V_{dd}.
- Adjust V_{B56} such that the V_{DS} of $M_{1,2}$ is 50 mV above their V_{DSAT}. Similarly, Adjust V_{B78} such that the V_{DS} of $M_{3,4}$ is 50 mV above their V_{DSAT}. i.e., M_{1-4} must be in saturation region with 50 mV margin.

Present the following results.

Tables: (Gain, transfer function etc. below refer to the differential quantities unless otherwise mentioned.)

(a) Specification table with your specific values and all the component values.

(b) Table showing simulation results: closed loop dc gain, closed loop 3-dB bandwidth, unity loop gain frequency, phase margin, rms output noise (integrated from 10 kHz to 100 MHz), fraction of noise variance (integrated from 10 kHz to 100 MHz) contributed by R_i, R_f, first stage, second stage, and R_L, Opamp open loop dc gain, positive and negative slew rates, positive and negative swing limits, HD_3, supply voltage, current consumption. Unity loop gain frequency and phase margin of the second stage CMFB loop. To find the opamp slew rates apply a large input step (from V_{cm} to $V_{cm} \pm V_{step}$) such that the first stage current is completely switched to one side.

To find HD_3, apply a sinusoidal input that results in a 1 V peak-peak output at a frequency that is 1/4th the bandwidth. Report HD_3, the ratio of the third harmonic to the fundamental (in dB).

Plots:

(a) Differential loop gain magnitude (dB) and phase (degrees). The unity loop gain frequency and phase margin must be marked. Break the loop as shown in Fig. 9.1(d) to simulate the loop gain. Use large L, C, e.g., $L = 10^6$ H, $C = 1$ F. The frequency range should be from $\sim 0.1\times$ dominant pole to where the loop gain is ~ -20 dB.

(b) Closed loop transfer function magnitude on a log-y scale showing the dc gain and the 3-dB bandwidth.

(c) Closed loop dc transfer curve. Vary V_i from -1 V to 1 V

(d) Small-signal step response: Output should step from 0 V to 0.1 V and back. Use a short rise time $\sim$ 100 ps.

(e) Large-signal step response: Output should step from 0 V to 1 V and back. Use a short rise time $\sim$ 100 ps. Get the slew rate from this. Check to see that the current is completely switched to one side in the first stage. Find the rising and falling slew rate from op and om.

(f) Output noise PSD and the input referred noise PSD of the closed loop amplifier from 10 kHz to 100 MHz.

(g) Input referred noise PSD of the opamp from 10 kHz to 100 MHz.

(h) Second stage CMFB loop gain. You can "break" the loop at the positive input of A_{cm2}. The unity loop gain frequency and phase margin must be marked.

(Not for submission):

- Remove the cascodes from the first stage, run the simulations and find out the differences
- Connect C_c to the drain of M_2 or M_4 instead of the drains of $M_{6,8}$. See if there is a difference in phase margin. You can also try connecting $C_c/2$ each to the drains of M_2 and M_4.

- What is pole frequency of the feedback network. This is due to the resistive divider with R_f and R_i in combination with the input capacitance of the opamp. You can take $C_{gs}/2$ to be the differential input capacitance. Does it matter in this design? If so, how would you mitigate its effect?

Assignment 10

Not for submission. You can try this out to get some experience with bandgap references and LDOs.

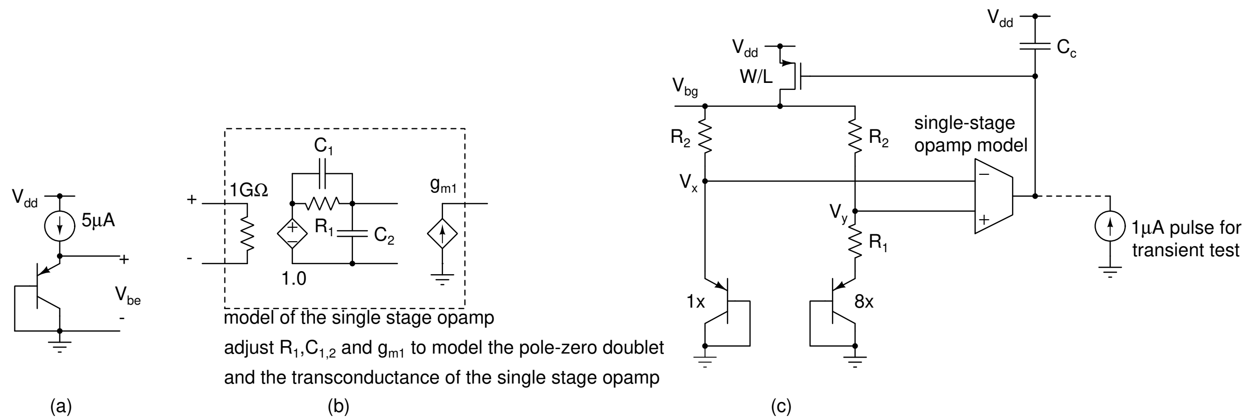


Figure 10.1: Problem 10.1. (a) Fully differential opamp with a pMOS input pair, (b) Fully differential opamp with a nMOS input pair, (c) Test circuit.

10.1. **Bandgap reference**: Bias a 1x sized diode connected PNP[4] at 5µA as shown in Fig. 10.1(a) and sweep the temperature from 0 to 100°C. Determine dV_{BE}/dT at 50°C.

Design the bandgap shown in Fig. 10.1(c). Choose R_1 for a quiescent current of 5µA and R_2 to get zero temperature coefficient at V_{bg}. Simulate the bandgap reference with the model of a single stage opamp assuming that the single stage opamp is made as in the earlier assignment. (Fig. 10.1(b)-model the gm, and the pole zero doublet). Choose C_c for ringing $\leq$ 10%. Test the bandgap reference by sweeping the temperature from 0 to 100°C and plot V_{bg}. Test the transient response by applying a 1 uA pulse to the output of the opamp. Adjust the values of R_1 and R_2 if necessary to get zero TC at 50°C.

Substitute the differential pair opamp designed in the previous assignments and simulate the temperature sensitivity of V_{bg} and the transient response to a current step at the output. Do this for the nominal supply voltage and ±10%.

10.2. **Low dropout voltage regulator**: A voltage regulator is nothing but a noninverting amplifier whose input is the bandgap voltage from a reference. In Fig. 10.2(a), the output voltage is $(R_2/R_1)V_{bg}$. By making R_2 variable, one can get a variable voltage output.

- The output impedance should be very low: This is accomplished by realizing a very high loop gain over as wide a bandwidth as possible.
- The efficiency $((V_{out}I_L)/(V_{dd}I_{sup}))$ should be very high: For this, the current $I_{sup} - I_L$ consumed by the circuit should be minimized (This makes it hard to satisfy the previous condition). The “dropout” $V_{dd} - V_{out}$ should be minimized.
- Usually only a positive I_L needs to be driven. The output voltage is constant over time. These are departures from conventional amplifiers.

Fig. 10.2(b) shows a “pass transistor” M_1 enclosed in a feedback loop. For simplicity, a unity gain case is shown. M_1 should have a high enough W/L to remain in saturation with the desired dropout and the highest output

[4] Use the model `ideal_pnp` in `ideal_diode.lib`

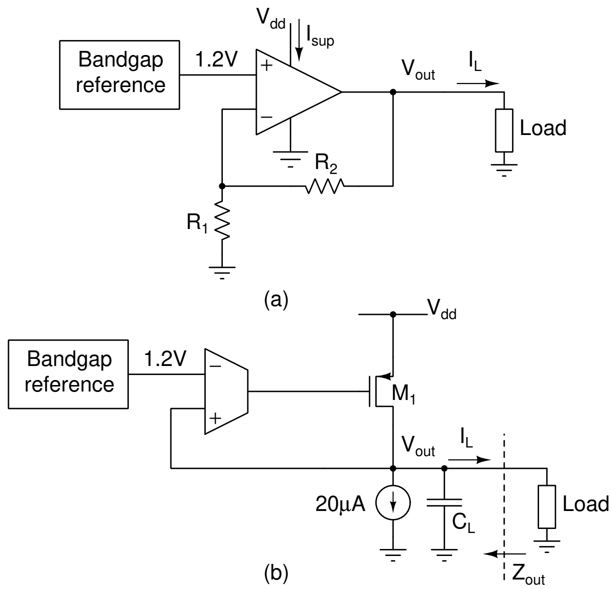


Figure 10.2: Low dropout regulator

current. Miller compensation around M_1 is usually not used because it severely compromises power supply rejection (Incremental voltage gain from V_{dd} to the output voltage).

Use the model in Fig. 10.1(b) for the single stage opamp. Use a 50µA quiescent current in M_1. Adjust the width (with minimum length) of M_1 for a dropout of 300 mV with a 50 mA current. You can use a 1.2V voltage source in place of the bandgap reference. Compensate the loop using a load capacitor C_L for a phase margin of 45° at $I_L = 0$ and $I_L = 50$ mA and choose the higher one. Do the following (except the last one) for two cases ($I_L = 0$ and $I_L = 50$ mA—you can use a current source for the load):

(a) Vary V_{dd} from 1.4 V to 1.8 V and plot V_{out}

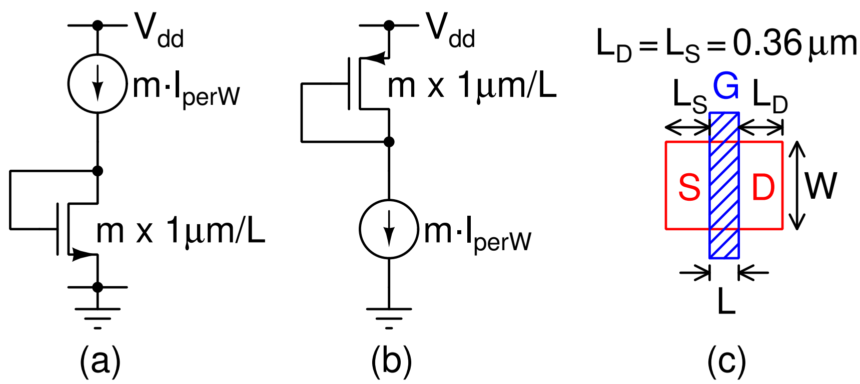
(b) Plot Z_{out} from 1 kHz to 10 MHz

(c) Plot the transfer function from V_{dd} to V_{out} from 1 kHz to 10 MHz

(d) Plot the small signal step response for a 10µA step in the output current

(e) Plot the large signal step response (I_L switching from zero to 1µA and 50 mA to zero)

Assignment: MOS characterization



(a) nMOS test circuit, (b) pMOS test circuit, (c) Unit transistor layout—A_d, A_s are the areas and P_d, P_s are the perimeters of the drain and source diffusion regions respectively. $A_d = WL_d$, $A_s = WL_s$, $P_d = 2(W + L_d)$, $P_s = 2(W + L_s)$.

Characterize the MOS transistors (file: `tsmc018.lib`) in a 0.18µm process using the circuits shown in Fig. **??**. Choose the transistor parameters and the current as follows.

Bias current	$10I_{perW}$
W	1µm
L	0.3µm, 0.4µm
$A_d = A_s$	0.36×10^{-12}m^2 (µm^2)
$P_d = P_s$	2.72×10^{-6}m (µm)
Multiplier m	10
V_{dd}	1.8 V

I_{perW} is the bias current per unit width. The transistor is 10 multiples of $1\mu\text{m}/L$ unit. Sweep I_{perW} in logarithmic steps from a small value (say 0.1 µA/µm) to 1 mA/µm and simulate the operating point. Set the temperature=100°C. This has to be repeated twice, once for each length given above. Plot the following versus I_{perW}:

- V_{GS}: Gate-source voltage. This is to find a "reasonable" operating region. i.e., if $V_{GS} = 1.5$ V with a 1.8 V supply, that operating point is likely unusable even if it is suitable from other criteria. While plotting, set the y-axis limits to (0, $0.75V_{dd}$).

- V_{DSAT}: Saturation voltage required across the device. Similar to the above, this is to find a "reasonable" operating region. If the current source requires 1 V across it, it is likely unusable with a 1.8 V supply. While plotting, set the y-axis limits to (0, $0.5V_{dd}$).

- $g_m/10$: g_m per unit transistor ($1\mu\text{m}/L$). This is to find the device sizing for a given g_m.

- g_m/g_{ds}: Intrinsic gain. This is to find the bias current density or device length to obtain a given dc gain.

- $f_T = f_{ug} = g_m/2\pi C_{gg}$: This is the intrinsic unity gain frequency and a measure of the speed that can be achieved. This is again for choosing the current density for tailoring the speed of the circuit.

In future designs, use the same transistors (W, L, A_d, A_s, P_d, P_s), but change the multiplier (m) to suit your requirements.

Assignment: Passive circuits

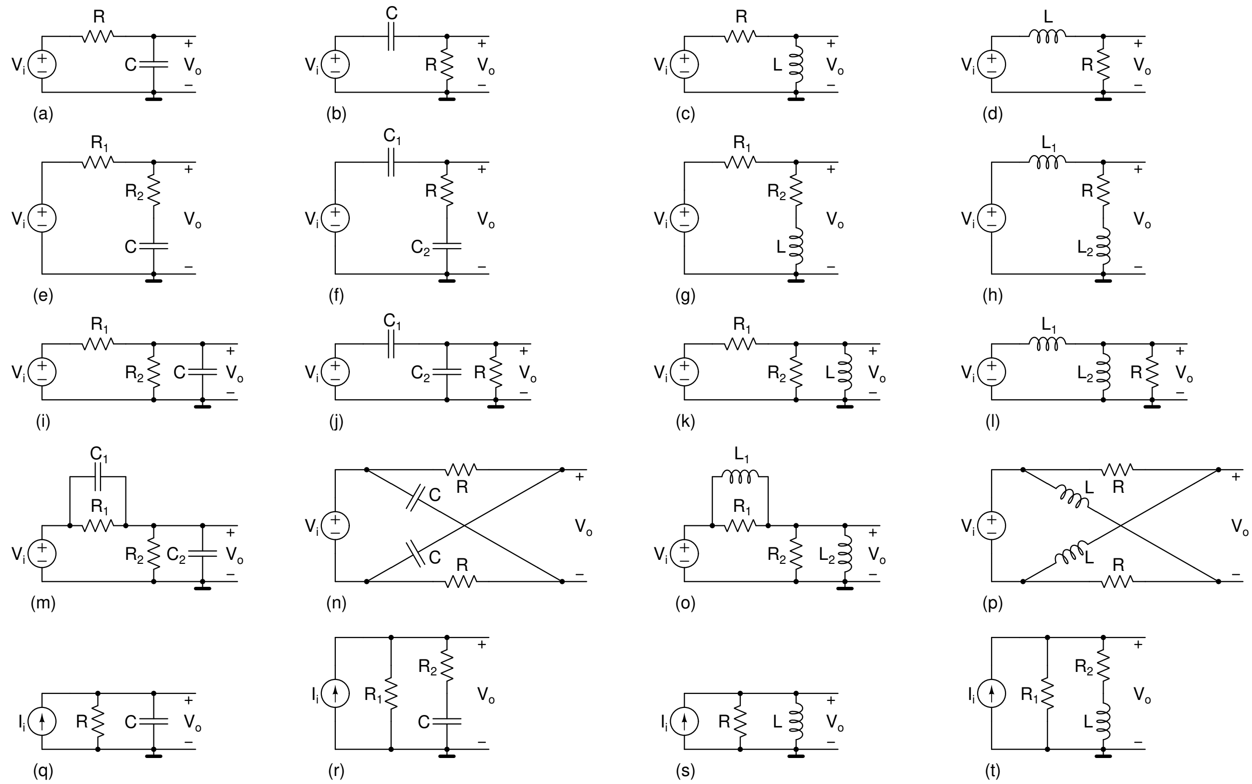


Figure 10.1: First-order circuits with R, L, and C.

10.1. Determine the following in Fig. 10.1. Try to get the answers intuitively by considering the electrical properties and limiting cases (high and low frequencies, $t = 0^+$, $t = \infty$, time constants, poles and zeros, null output condition etc.). Compare them to the formally derived answers and improve your intuition if necessary. Units 9, 10, 11 of the course `https://nptel.ac.in/courses/108106172` describe these concepts.

(a) Bode magnitude and phase plots of $V_o(j\omega)/V_i(j\omega)$ or $V_o(j\omega)/I_i(j\omega)$.

(b) Transfer function $V_o(s)/V_i(s)$ or $V_o(s)/I_i(s)$.

(c) Values of $V_o(t)$ and dV_o/dt at $t = 0^+$ and $t = \infty$ with a unit step input. You should be able to do this without explicitly solving for $V_o(t)$ by considering currents and voltages in the circuit.

(d) Values of $V_o(t)$ and dV_o/dt at $t = 0^+$ and $t = \infty$ with a unit impulse input. You should be able to do this without explicitly solving for $V_o(t)$ by considering currents and voltages in the circuit.

(e) Expression for the unit step response.

(f) You can attempt all of the above for other branch voltages and currents instead of v_o.

10.2. Determine the following in Fig. 10.2. Try to get the answers intuitively by considering the electrical properties and limiting cases (high and low frequencies, $t = 0^+$, $t = \infty$, natural frequency and quality/damping factors, null output condition etc.). For RLC circuits first identify whether they are series or parallel resonance circuits. Each of them has a characteristic denominator. Compare them to the formally derived answers and improve your intuition if necessary.

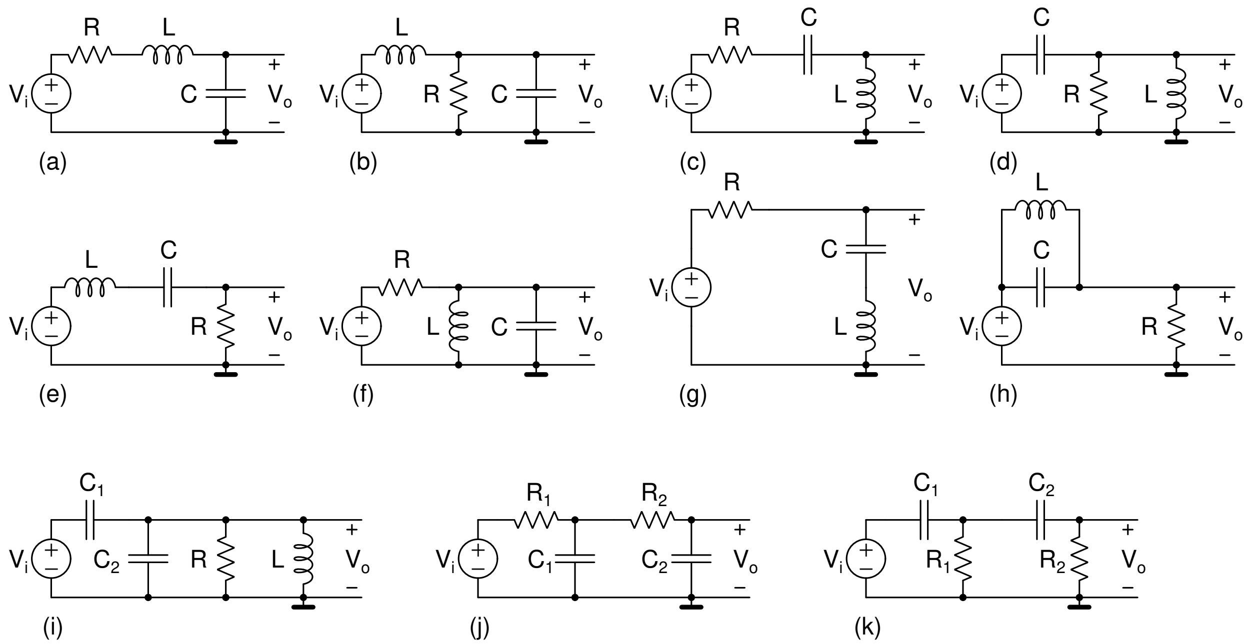


Figure 10.2: Second-order circuits with R, L, and C.

(a) Transfer function $V_o(s)/V_i(s)$ or $V_o(s)/I_i(s)$.

(b) Quality factor or damping factor of the above transfer functions.

(c) Bode magnitude and phase plots of $V_o(j\omega)/V_i(j\omega)$ or $V_o(j\omega)/I_i(j\omega)$.

(d) Values of $V_o(t)$ and dV_o/dt at $t = 0^+$ and $t = \infty$ with a unit step input. You should be able to do this without explicitly solving for $V_o(t)$ by considering currents and voltages in the circuit.

(e) Values of $V_o(t)$ and dV_o/dt at $t = 0^+$ and $t = \infty$ with a unit impulse input. You should be able to do this without explicitly solving for $V_o(t)$ by considering currents and voltages in the circuit.

(f) In Fig. 10.2(j, k), what is the highest quality factor (or the lowest damping factor) achievable? What are the necessary constraints?

(g) Expression for the unit step response.

(h) You can attempt all of the above for other branch voltages and currents instead of v_o.